Théo Dumont

PhD student at LIGM, Univ. Gustave Eiffel (Marne-la-Vallée, France)
Optimal transport / Riemannian geometry / Machine learning

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Overview

1.1. **Summary.** I am a PhD student in mathematics at LIGM (Université Gustave Eiffel), under the supervision of François-Xavier Vialard, Théo Lacombe and Virginie Ehrlacher. I am studying the geometry of several optimizations problems related to optimal transport (OT), such as regularized OT or Gromov–Wasserstein.

1.2. **Interests and recent work.** My interests span the theory of OT and infinite-dimensional Riemannian geometry, and I am passionate about the interplay between those fields. In particular, I am fond of gradient flows and functional inequalities when they both arise from various choices of metrics on the space of probability measures. I recently worked with F.X. Vialard and T. Lacombe on gradient flows constrained to the space of OT maps [1] and on the Gromov–Wasserstein problem [2], with P. Harms on the infinite-dimensional geometry of neural networks, and with K. Modin on gradient flows in the diffeomorphism group. I also have a strong background in computer science and machine learning, both theoretical and applied, thanks to my academic background and my early research experiences [3, 4].

I also love teaching—whether it is mathematics, computer science, or societal topics. I love creating visuals, summarizing dense content in a clear way, and thinking about the best way to make something easily understandable.

Curriculum

2.1. **Education.**

2023– : **PhD in mathematics, LIGM (Université Gustave Eiffel)**
Thesis: *Geometry of some optimization problems on spaces of measures* [1]
Supervisors: François-Xavier Vialard, Théo Lacombe and Virginie Ehrlacher

2021–2022: **M.Sc., ENS Paris-Saclay**, Master 2 MVA (*Mathematics, Computer Vision and Machine Learning*)
Thesis: *Existence of Monge maps for the Gromov–Wasserstein problem* (link of the thesis) [2]
Supervisors: François-Xavier Vialard and Théo Lacombe

2018–2022: **M.Sc., Mines Paris – PSL University** (*Applied Mathematics and Computer Science*) [3, 5, 4]

2016–2018: **B.Sc., Lycée Sainte-Geneviève** (*Mathematics, Physics, Fundamental and applied Computer Science*)

2.2. **Teaching experiences.**

- **Mathematics**

(3 years) 2023–2026	*Computer Science for Mathematics*, Year 2, Université Gustave Eiffel, France.
2019–2025	*Private tutor in Mathematics*, Years 1,2,3.

- **Machine Learning and Computer Science**

(2 years) 2023–2025	*Algorithm and data structures*, Year 1, Université Gustave Eiffel, France.
(2 years) 2024–2026	*Machine learning*, Year 4, Université Gustave Eiffel, France.
(2 years) 2023–2025	*Data science*, Year 3, Mines Paris, France (link).

- **Ecological crisis**

(2 years) 2024–2026	*Climate change*, Year 1, Université Gustave Eiffel, France.

2.3. **Research experiences.** Here is a list of my research stays prior to my PhD.

2023: (2 months) **Chalmers University (Göteborg, Sweden)**, internship (infinite-dimensional Riemannian geometry)
Gradient flows in the diffeomorphism group
Supervisor: Klas Modin

2022–2023: (6 months) **NTU Singapore**, internship (infinite-dimensional Riemannian geometry)
Intrinsic geometry of neural networks
Supervisor: Philipp Harms

2022: (6 months) **Université Gustave Eiffel – LIGM**, internship (optimal transport)
Existence of Monge maps for Gromov–Wasserstein [2]
Supervisors: François-Xavier Vialard and Théo Lacombe

2021: (6 months) **Tekal.ai, working with MIT CSAIL**, internship (deep learning)
Video memorability prediction [3]
Supervisor: Camilo Fosco

2020–2021: (6 months) **InterDigital, Inc.**, internship (deep learning)
Temporal segmentation of facial features
Supervisors: Claire-Hélène Demarty and Frédéric Lefebvre

2020: (3 months) **Turing Centre for Living Systems (INMED)**, internship (computer vision)
Image processing and dimensionality reduction for neuronal coactivity detection in the brain [5]
Supervisors: Rosa Cossart and Julien Denis

2019–2020: (6 months) **Mines Paris – Center for Mathematical Morphology**, internship (deep learning)
Image segmentation by superpixels [4]
Supervisor: Bruno Figliuzzi

Publication list

See also my Google Scholar page.

3.1. Optimal transport and geometry.

[1] T. Dumont, T. Lacombe, and F.-X. Vialard. "Learning Monge maps by lifting and constraining Wasserstein gradient flows". In: *to appear* (2026). POSTER.

[2] T. Dumont, T. Lacombe, and F.-X. Vialard. "On the existence of Monge maps for the Gromov–Wasserstein problem". In: *Foundations of Computational Mathematics* (2024), pp. 1–48. PDF CODE SLIDES POSTER.

3.2. Applied ML and image analysis.

[3] T. Dumont, J. S. Hevia, and C. L. Fosco. "Modular Memorability: Tiered Representations for Video Memorability Prediction". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2023, pp. 10751–10760. PDF CODE.

[4] D. Paulovics, B. Figliuzzi, T. Dumont, and F. Blanc. "A supervised algorithm entirely trained on a synthetic dataset to segment granular suspension images". Preprint. 2023. PDF CODE.

[5] R. F. Dard, E. Leprince, J. Denis, S. R. Balappa, D. Suchkov, R. Boyce, C. Lopez, M. Giorgi-Kurz, T. Szwagier, T. Dumont, et al. "The rapid developmental rise of somatic inhibition disengages hippocampal dynamics from self-motion". In: *Elife* 11 (2022), e78116. PDF CODE.

Talks and posters

Here are listed talks on my original research (●), talks on other people's work (○), and posters (☆).

— **Learning Monge maps by lifting and constraining Wasserstein gradient flow** [1] POSTER

- ● *June 2026* | *Curves and Surfaces*, Saint-Malo, France.
- ☆ *Oct 2025* | *GdR IASIS: "Modèles génératifs : diffusion, flow matching et leurs applications"*, ENS Lyon, France.
- ● *Jan 2025* | *Infinite-dimensional Geometry: Theory and Applications*, ESI Vienna, Austria.

— **Existence of Monge maps for the Gromov–Wasserstein problem** [2] SLIDES POSTER

- ● *Oct 2024* | *Congrès des Jeunes Chercheur.e.s en Mathématiques Appliquées (CJC-MA)*, ENS Lyon, France.
- ☆ *Aug 2024* | *Machine Learning in Infinite Dimensions*, Bath, UK.
- ☆ *May 2024* | *Geometric Sciences in Action: from geometric statistics to shape analysis*, CIRM, Marseille, France, awarded the first place for the poster competition.
- ● *Jan 2024* | *MAP5 lab PhDs' seminar*, Université Paris Cité, France.
- ● *Oct 2023* | *Oberwolfach seminar: "Variational and information flows in ML and OT"*, Oberwolfach, Germany.

— **Logarithmic Sobolev inequalities and related topics** NOTES

- ○ *Oct 2024* | *New Monge problems*, Paris, France.

— **Infinite-dimensional Riemannian geometry of optimal transport and Gromov–Wasserstein**

- ● *Nov 2023* | *Shape analysis seminar*, INRIA Paris, France.

Summer schools, conferences, workshops and seminars

Here is a list of my participation in events, with indication of whether I presented a talk on my original research (●), a talk on other people's work (○), or a poster (☆).

— **Summer schools**

Sept 2024 | *Metric Measure Spaces, Ricci Curvature, and Optimal Transport*, Lake Como, Italy.
July 2024 | *Festum Pi: "Modern research in analysis"*, Chania, Greece.
● *Oct 2023* | *Oberwolfach seminar: "Variational and information flows in ML and OT"*, Oberwolfach, Germany.
Jun 2022 | *An optimization perspective on sampling using optimal transport* (my notes), Paris, France.

— **Conferences**

● *Jan 2025* | *Infinite-dimensional Geometry: Theory and Applications*, ESI Vienna, Austria.
☆ *Aug 2024* | *Machine Learning in Infinite Dimensions*, Bath, UK.
☆ *May 2024* | *Geometric Sciences in Action: from geometric statistics to shape analysis*, CIRM, Marseille, France, awarded the first place for the poster competition.
Aug 2023 | *Geometric Science of Information*, Saint-Malo, France.
Oct 2022 | *Geometry, Topology and Statistics in Data Sciences*, IHP, Paris, France.

— **Workshops**

☆ *Oct 2025* | *GdR IASIS: "Modèles génératifs : diffusion, flow matching et leurs applications"*, ENS Lyon, France.
July 2025 | *Labos 1point5: Enseigner les transitions écologiques et sociales dans le supérieur*, Université Lyon 1, France.
June 2025 | *Rencontres Archipel*, Université Lyon 1, France.
● *Oct 2024* | *Congrès des Jeunes Chercheur.e.s en Mathématiques Appliquées (CJC-MA)*, ENS Lyon, France.

— **Seminars**

○ *Oct 2024* | *New Monge problems*, Paris, France.
● *Jan 2024* | *MAP5 lab PhDs' seminar*, Université Paris Cité, France.
● *Nov 2023* | *Shape analysis seminar*, INRIA Paris, France.

Computer and Language skills

Computer skills

Proficient:	Python, PyTorch, git, Unix, LaTeX, BASH, SQL, Tikz, Adobe Suite
Intermediate:	Java, OCaml, HTML, CSS, Coq

See also my GitHub, `theodumont`.

Language skills

English:	Fluent	(C1, TOEFL 113/120)
French:	Native speaker	
Italian:	Intermediate	(B1)
Japanese:	Basic	(A2)